

Ans 1 Let X be 6-sided die
and Y be 10-sided die

There are $10 \times 6 = 60$ equally
likely outcomes

1.1 Probability that $Y = b$,
given different numbers

There are 6 matching outcomes
 $(1,1), \dots, (6,6)$

thus $60 - 6 = 54$ outcomes
have different numbers

of these, 5 have $Y = b : (1,b), \dots, (5,b)$

$$P(Y = b \mid X \neq Y) = \frac{5}{54}$$

1.2 The outcome with $\text{Sum} = 10$
and $X = 6 \Rightarrow (6,4)$

$$P(X = 6 \mid X + Y = 10) = \frac{1}{6}$$

Ans 2 Let B_j denotes the event that the ball is in box j and F_i denote the event that searching box i fails to find it

The probability of finding the ball in box i is $\alpha_i p_i$

$$P(F_i) = 1 - \alpha_i p_i$$

By bayes theorem,

$$P(B_j | F_i) = \frac{P(F_i | B_j) P(B_j)}{P(F_i)}$$

i if $j \neq i$: the ball is in another box, so searching box i fails, thus $P(F_i | B_j) = 1$

ii if $j = i$: the ball is in searched box but the search misses it with probability $1 - \alpha_i$

Therefore,

$$P(B_j | F_i) = \begin{cases} \frac{p_j}{1 - \alpha_i p_i} & j \neq i \\ \frac{(1 - \alpha_i) p_i}{1 - \alpha_i p_i} & j = i \end{cases}$$

This assumes $1 - \alpha_i p_i > 0$,
So the conditioning has positive probability

Ans 3 Let K be the number of black balls transferred from B to A . After the transfer urn A contains 11 balls

3.1 Probability of drawing a black ball from A

Expected number of black balls transferred is

$$E[K] = 2 \left(\frac{9}{13} \right) = \frac{18}{13}$$

Therefore,

$$P(\text{black}) = \frac{3 + E[K]}{11}$$

$$= \frac{3 + \frac{18}{13}}{11}$$

$$= \frac{57}{143}$$

3.2 Probability both transferred balls were black, given a white draw

Let T mean both transferred balls were black and W mean ball drawn from A was white

$$P(T) = \frac{\binom{9}{2}}{\binom{13}{2}}$$

$$= \frac{6}{13}$$

$$P(W|T) = \frac{6}{11}$$

From part 1,

$$P(W) = \frac{1-57}{143} = \frac{86}{143}$$

Using Bayes theorem,

$$P(T|W) = \frac{P(W|T)P(T)}{P(W)}$$

$$= \frac{6}{11} \times \frac{6}{13} = \frac{18}{43}$$

Ans 4 Current Total = $4+10=14$

So we need 4 more points
drawing at most 2 additional
Card

There are $39-7=32$ unseen card

Drawing next card is equally
likely to be any of 32

1 Reach 18 with 1 additional card

So we have to draw a 4

$$P(\text{draw a 4}) = \frac{2}{32}$$

2 Reach 18 with 2 additional Card : $[(A, 3), (3, A), (2, 2)]$

with $A = 1$

$$P(\text{two-card Success}) =$$

$$\frac{3}{32} \times \frac{3}{31} + \frac{3}{32} \times \frac{3}{31} + \frac{3}{32} \times \frac{2}{31}$$

$$= \frac{24}{992}$$

Adding these Mutually Exclusive possibility

$$P = \frac{2}{32} + \frac{24}{992}$$

$$= \frac{43}{496} \approx 8.67\%$$

Ans 5.1 Independent Events

By De Morgan law, the complement of 'at least one event occurs' is 'none of event occurs'

Therefore,

$$\begin{aligned} P\left(\bigcup_{i=1}^n E_i\right) &= 1 - \prod_{i=1}^n P(E_i^c) \\ &= 1 - \prod_{i=1}^n (1 - P(E_i)) \end{aligned}$$

5.2 Mutually Exclusive Events

Let $U = \bigcup_{i=1}^{\infty} E_i$, Since $E_j \subseteq U$,

we have $E_j \cap U = E_j$.

By definition of Conditional Probability

$$P(\epsilon_j | U) = \frac{P(\epsilon_j \cap U)}{P(U)}$$

$$= \frac{P(\epsilon_j)}{P(U)}$$

Because the events are mutually exclusive, the countable additivity give

$$P(U) = \sum_{i=1}^{\infty} P(\epsilon_i) > 0$$

Hence

$$P\left(\epsilon_j \mid \bigcup_{i=1}^{\infty} \epsilon_i\right) = \frac{P(\epsilon_j)}{\sum_{i=1}^{\infty} P(\epsilon_i)}$$

Ans 6 The equally likely outcome are

$$\Omega = \{HH, HT, TH, TT\}$$

Thus,

$$A = \{HH, HT\}, \quad B = \{HH, TH\},$$

$$C = \{HH, TT\}$$

$$\text{So, } P(A) = P(B) = P(C) = \frac{1}{2}$$

i Pairwise Independence

$$\text{Since } A \cap B = A \cap C = B \cap C = \{HH\},$$

$$P(A \cap B) = P(A \cap C) = P(B \cap C) = \frac{1}{4}$$

Each equals the product of corresponding probabilities

$$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Therefore A, B, C are pairwise independent

6.2 Mutual Independence

$$P(A \cap B \cap C) = P(\{HH\}) = \frac{1}{4}$$

but

$$P(A) \cdot P(B) \cdot P(C) = \frac{1}{8}$$

Hence the events are not mutually independent

6.3 Given C , the equally likely outcomes are

HH and TT. Therefore

$$P(A|C) = P(B|C) = \frac{1}{2}$$

$$P(A \cap B | C) = \frac{1}{2}$$

However

$$P(A|C) \cdot P(B|C) = \frac{1}{4}$$

Since $\frac{1}{4} \neq \frac{1}{2}$

Thus A and B are not conditionally independent given C. Knowing the coin match means that if the first is heads and second must also be heads